From Initial Beliefs to Better Forecasts: Auditing Recurrent Linguistic Evaluation of AI Systems

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Abstract

Predicting whether an AI system will answer a particular question correctly requires calibrated, item-conditioned forecasts under sparse acquired labels and benchmark shift. We audit the organizer-provided Bayesian Linguistic Evaluator (BLE), which elicits an initial language-model belief, retrieves admissible historical measurements, and revises that belief through bounded actions. We study rather than claim invention of this architecture. In a frozen, outcome-sealed public confirmation on 32 new questions from four previously encountered benchmark families at budgets 0 and 7, GPT-6 Luna BLE R1 achieves **0.153728** Brier versus **0.203883** for direct Luna prediction (a paired reduction of 0.050155; source-cluster bootstrap 95% interval [0.001569, 0.103059]). The advantage separates descriptively into 0.031767 before the first retrieval and 0.018388 during the subsequent recurrent phase; neither term identifies retrieval's causal effect. We then document a distinct, successfully scored OpenAI Responses API implementation, BLE 1.5-A, with official formative Brier ALC **0.1590607804** across nine subject–benchmark groups, 489 distinct evaluated pairs, and all six acquisition budgets. Mean Brier decreases from 0.171907 at zero labels to 0.145128 at 31 labels. The public diagnostic and hosted ALC use different evaluation sets and cannot be compared as paired outcomes. Our contribution is an auditable experimental and deployment account: explicit belief states, evidence controls, transport validation, reproducibility boundaries, and the limits of inference under distribution shift.

1 Introduction

Forecasting an AI system's success is not equivalent to estimating its average benchmark accuracy. Outcomes depend on the task item, system configuration, harness, reasoning effort, and availability of limited observations. Two systems with similar mean accuracy may fail on different questions; the required prediction is an item-conditioned probability, not a single system score.

The Predictive AI Evaluation Competition (PAIEC) supplies the system's visible attributes, an evaluation item, and a limited set of acquired outcomes while withholding the target response. Its metric macro-averages Brier area under the label-acquisition learning curve (ALC) across subject–benchmark groups [1]. For target system s and item q, our object is

$$p_{s,q} = \Pr(Y_{s,q} = 1 \mid a_s, x_q, \mathcal{L}_s, \mathcal{D}_{\text{pub}}), \qquad Y_{s,q} \in \{0, 1\}. \tag{1}$$

where a_s is the permitted system configuration, x_q the item, $\mathcal{L}_s$ the acquired labels, and $\mathcal{D}_{\text{pub}}$ the admissible historical store.

The competition baseline includes a Bayesian Linguistic Evaluator (BLE): a prompt-and-tool procedure that expresses a probabilistic belief, queries historical evidence, and revises a forecast [2]. We treat this as an auditable linguistic measurement-state process, not an altered Transformer or a

Preprint.

learned Bayesian filter. This framing is conceptually related to the study of recurrent computation [3], without sharing the neural recurrence of a Recurrent Looped Transformer.

Contributions. We provide (i) an outcome-sealed, paired comparison between direct Luna and BLE R1 with explicit separation of initial and later revisions; (ii) a separately reported official six-budget hosted result from the deployed BLE 1.5-A runtime; (iii) a precise account of allowed evidence, the JSON-action inference transport, and deployment reliability; and (iv) negative transfer and calibration limitations that delimit the claims. We do not attribute the organizer's BLE method to ourselves, infer unseen hidden-family identities, or treat two different scoring protocols as interchangeable.

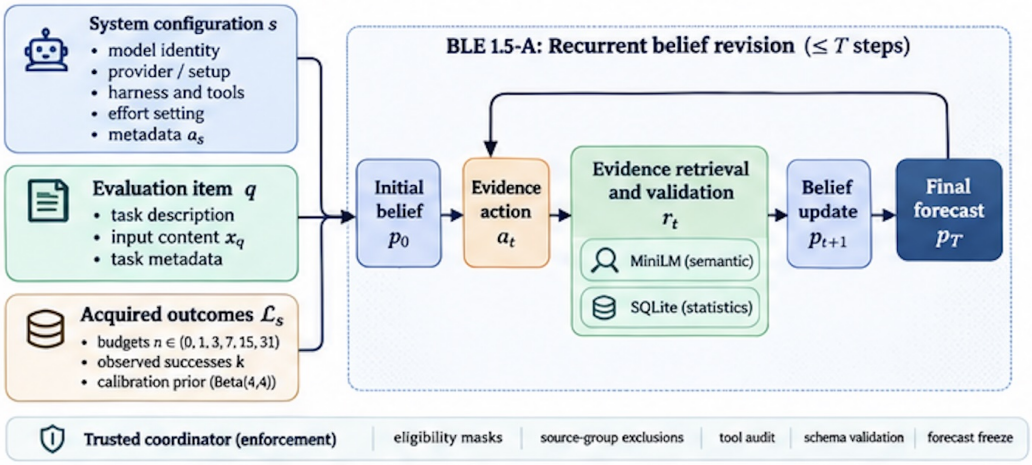


Figure 1. Audited BLE 1.5-A forecasting pathway. Target system configuration, task item, and acquired labels condition an initial GPT-6 Luna Medium belief. The coordinator executes bounded JSON actions over permitted MiniLM and SQLite evidence; Luna revises its belief and returns a final probability. This is linguistic recurrence, not neural KV-cache recurrence.

2 Method: an auditable recurrent measurement process

2.1 Belief state and bounded update

An observable inference state at step t is

$$S_t = (x, \mathcal{E}_t, b_t, h_t), \qquad x = (a_s, x_q, \mathcal{L}_s), \qquad b_t = (p_t, c_t, e_t^+, e_t^-, u_t), \tag{2}$$

with accumulated admissible evidence $\mathcal{E}_t$, reasoning/action history h_t , and an expressed belief containing probability p_t , confidence, supporting and opposing considerations, and unresolved questions. An action a_t followed by validated tool result r_t induces a bounded model/coordinator transition,

$$S_{t+1} = F_\theta(S_t, a_t, r_t), \qquad p_{t+1} = \pi(S_{t+1}). \tag{3}$$

The map F_θ is not an exact Bayesian posterior update and is not trained as a state-transition model. The sequence of expressed probabilities is available for auditing. Additional retrieval can help, have no relevance, or cause an inappropriate revision.

2.2 Admissibility, relevance, and acquired labels

An evidence coordinator restricts access to historical measurements, source groups, variants, and unavailable responses. It also distinguishes permitted records from relevant records: a high measured score for another model/harness/task need not transfer to the target item. Comparability of system, harness, task and provenance is evaluated linguistically, not through a separately trained relevance model.

Acquired binary labels are governed by $n \in \{0, 1, 3, 7, 15, 31\}$ in the official setting. Our reference Beta(4, 4) posterior mean is

$$p_\beta(k, n) = \frac{4 + k}{8 + n}, \tag{4}$$

for k acquired successes in n trials. This is an acquisition-only reference; BLE additionally reasons about the target item and retrieved evidence. The exact admissibility and acquisition code in the submitted artifact governs the official result, not a conceptual diagram.

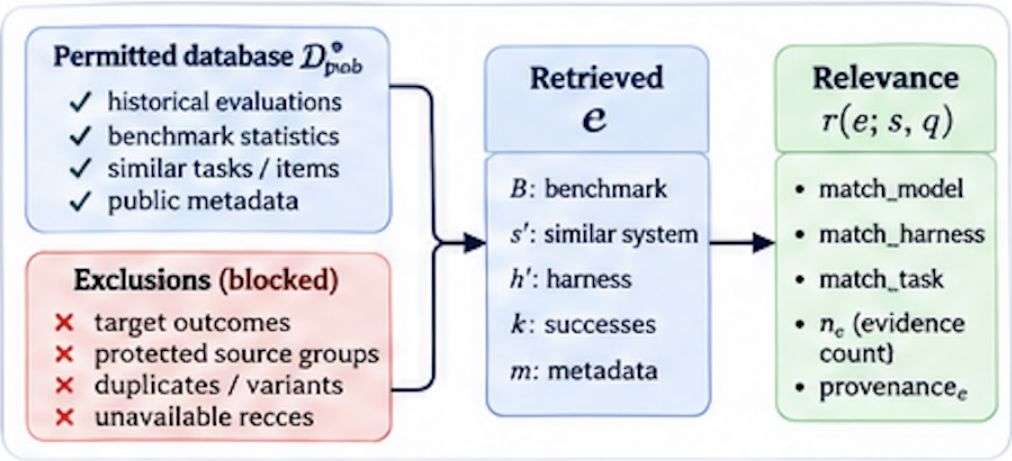


Figure 2. Evidence admissibility is distinct from relevance. Coordinator rules prevent retrieval of protected target outcomes and source variants; allowed measurements still require judgments about model, harness, item and provenance comparability. Passing an eligibility check is not evidence of predictive utility.

2.3 Why a linguistic revision may help or hurt

For an ideal conditional mean $p_t^\star = \mathbb{E}[Y \mid \mathcal{F}_t]$ and nested information $\mathcal{F}_t \subseteq \mathcal{F}_{t+1}$, squared-error projection gives

$$\mathbb{E}[(Y - p_t^\star)^2] - \mathbb{E}[(Y - p_{t+1}^\star)^2] = \mathbb{E}[(p_{t+1}^\star - p_t^\star)^2] \geq 0. \tag{5}$$

For an approximate linguistic update $p \mapsto p + \delta$, however, realized loss changes by $2\delta(y - p) - \delta^2$, which may be negative. The recurrent procedure therefore warrants empirical evaluation, not an assumption that more tool calls always improve calibration. We use the strictly proper Brier and negative log-likelihood scores [4, 5].

3 From research BLE R1 to hosted BLE 1.5-A

3.1 Two execution transports, two evidentiary roles

The historical R1 development and confirmation ran in ChatGPT-authenticated GPT-6 Luna Medium Codex CLI sessions, with a trusted research coordinator and outcome sealing. The separate, later production artifact BLE 1.5-A used GPT-6 Luna Medium through the paid OpenAI Responses API, running the official prediction interface under Codabench. The hosted artifact preserves the recurrent, evidence-grounded intent but is not assumed bitwise equivalent to the historical CLI transport; its particular prompts, action bounds and error handling must be read from the frozen submitted program. No cross-transport matched-output test is claimed here.

An implementation distinction matters: the scored A runtime requests structured JSON-coded BLE actions, with `tools: []` at the Responses API layer; its Python coordinator dispatches the actions. It does not rely on the API's native `function_call` items or on an OpenAI Agents SDK orchestration loop. Retrieval uses locally packaged MiniLM semantic similarity and SQLite measurement queries, constrained by source/target exclusions. The model can emit a terminal submission or an

interim belief when a later recoverable failure prevents further progress, according to the frozen implementation policy.

3.2 Failure containment and execution boundaries

Hosted inference requires more than a valid prompt: model assets must initialize, retrieval must honor eligibility masks, Responses continuations and JSON payloads must validate, and an interrupted interaction must not propagate a batch-fatal exception when a prior valid model belief can be retained. Relevant failure classes include response timeouts, provider throttling, malformed outputs, context limits, and financial reservation exhaustion. A historical development failure conflated oversized unsent context with a financial budget error; the frozen A variant was subsequently audited for its error classifications and preservation of earlier model-derived probabilities.

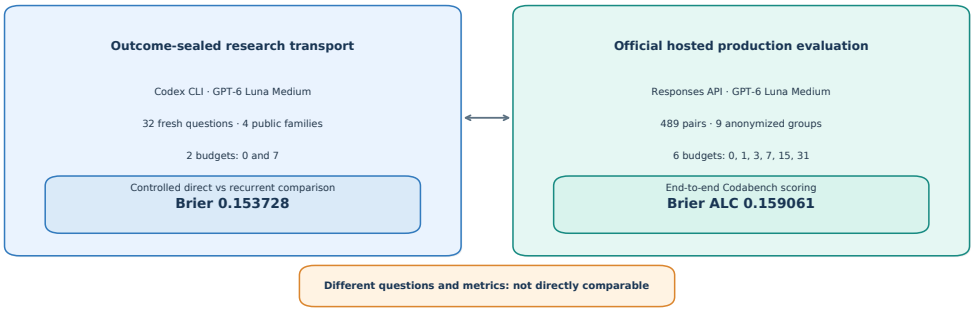


Figure 3. Two distinct evaluation tracks. Public R1 confirmation used a 32-question, two-budget outcome-sealed panel through Codex CLI. The later production BLE 1.5-A result was generated through the Responses API and officially scored across nine groups and six budgets. Brier 0.153728 and Brier ALC 0.159061 come from different samples and aggregations; their numerical proximity is not a paired generalization estimate.

The official BLE 1.5-A run, submission #1032237, reached *Finished* with complete scoring on all groups and budgets. This establishes a functioning hosted end-to-end path. It does not disclose the organizer-counted execution duration, so a visible prolonged *Running* status does not establish or refute compliance with the rules' eight-hour execution limit. Separate stress testing of the later A.2 release is not evidence of A's real-world fault probability (Appendix F).

4 Experimental design and scoring

4.1 Public outcome-sealed research comparison

The historical development stage compared R1 and R2 on 16 questions from four training families at $n \in \{0, 7\}$. Because development outcomes were previously revealed, this was method selection, not independent confirmation. R1 was then frozen and evaluated on 32 fresh, source-group-disjoint questions (eight each from LexEval, LiveAoPSBench, MathArena and MedAgentBoard) at those two budgets, producing 64 matched question–budget units per arm. The four families had been encountered during research; the questions were fresh, but this is not family-held-out transfer.

Direct Luna (A0), an initial BLE R1 belief, final BLE R1, and a Beta(4, 4) reference were scored on common underlying outcomes. Outcome access was sealed until accepted predictions were frozen; acquired labels and protected variants were excluded from scored target outcomes. All 146 retrieval packets passed archived source/protection checks. Predictions from different budgets on the same item are dependent, so intervals use the recorded source-group cluster bootstrap rather than treating 64 units as 64 independent questions.

For a binary outcome, $\ell_{\mathrm{Brier}}(p, y) = (p - y)^2$ and $\ell_{\mathrm{NLL}}(p, y) = -y \log p - (1 - y) \log(1 - p)$. The public experiment uses the scorer's equal-family, equal-budget diagnostic aggregation and is not a six-budget official ALC estimate.

4.2 Official formative six-budget scoring

The official competition computes within-group Brier B_n at budgets $n \in \{0, 1, 3, 7, 15, 31\}$ and forms

$$\begin{aligned} \mathrm{ALC}_g &= 0.1B_{g,0} + 0.2B_{g,1} + 0.2B_{g,3} + 0.2B_{g,7} + 0.2B_{g,15} + 0.1B_{g,31}, \\ \mathrm{ALC}_{\mathrm{macro}} &= \frac{1}{G}\sum_{g=1}^{G} \mathrm{ALC}_g. \end{aligned} \tag{6}$$

For BLE 1.5-A, $G = 9$ subject–benchmark groups containing 489 distinct pairs. Each of the six acquisition settings was scored for every group. This is a formative held-back evaluation with group-level summary feedback, not an independently sampled paired comparison against direct Luna; no private target outcomes were available to us for a new post-hoc ablation.

5 Results

5.1 Controlled public R1 confirmation

Table 1. Outcome-sealed public confirmation. Thirty-two fresh questions, four previously encountered families, budgets 0 and 7. Diagnostic Brier and NLL are not official six-budget ALC. Lower is better.

Method	Brier ↓	NLL ↓
Beta(4,4)	0.221448	0.634511
Direct Luna (A0)	0.203883	0.581099
R1 initial belief	0.172116	0.511191
R1 final BLE	**0.153728**	**0.471989**
Frozen 50/50 hybrid	0.195913	0.575572

Final R1 reduces Brier relative to direct Luna by **0.050155** (24.6% relative); the recorded source-cluster 95% bootstrap interval is [0.001569, 0.103059]. Both Brier and NLL improve. The arithmetic decomposition is

$$\Delta_{\mathrm{total}} = \underbrace{[B(p_{\mathrm{A0}}) - B(p_{\mathrm{R1},0})]}_{\Delta_{\mathrm{context}}=0.031767} + \underbrace{[B(p_{\mathrm{R1},0}) - B(p_{\mathrm{R1},T})]}_{\Delta_{\mathrm{revision}}=0.018388}. \tag{7}$$

Because the prompts differ before the initial belief, and recurrence includes both tool evidence and additional deliberation, neither term is a causal estimate of retrieval's benefit. R1 improved in three of four public families and slightly worsened in MedAgentBoard; detailed family scores are in Appendix C.

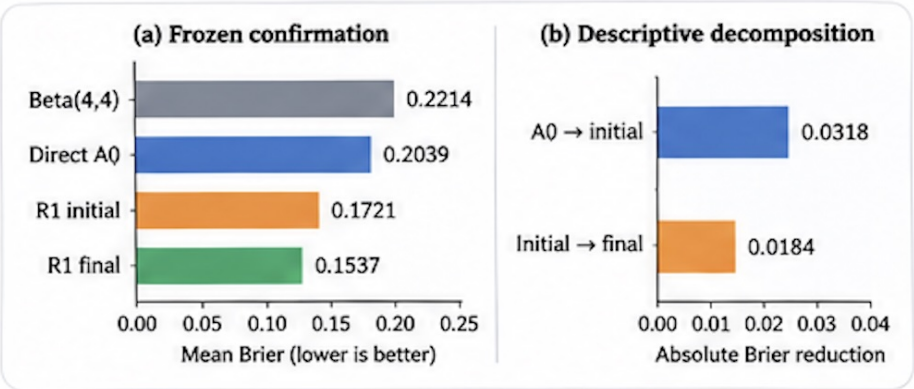


Figure 4. Public confirmation and descriptive decomposition. Final R1 obtains Brier 0.153728 versus 0.203883 for direct Luna and 0.221448 for Beta(4,4). Of the 0.050155 paired gain, 0.031767 precedes retrieval and 0.018388 follows in the recurrent phase. Different prompting and further reasoning preclude a retrieval-specific causal interpretation.

5.2 Official hosted BLE 1.5-A formative performance

The Codabench scorer reports **Brier ALC 0.1590607804** for submission #1032237. All nine groups received scores at all six budgets, corresponding to 489 distinct subject–item pairs. The equal-group macro mean reconstructed from the rounded group table is 0.1590607; the small difference from the displayed official value is rounding, not a separate score.

Table 2. Hosted BLE 1.5-A mean Brier by acquisition budget. Each column is macro-averaged across nine scored groups. Applying the official weights to these six-decimal Brier values gives ALC 0.1590607. The official ALC is 0.1590607804.

B_0	B_1	B_3	B_7	B_{15}	B_{31}
0.171907	0.169699	0.162202	0.157308	0.147577	0.145128

The mean Brier falls by 0.026779 (15.6% relative to B_0) from budget 0 to 31. This is a descriptive within-submission association with budget, not evidence that this particular acquisition policy beats random or that label acquisition caused the entire reduction. The ALC is 36.4% below the uninformative constant-0.5 Brier loss of 0.25, a reference rather than a competitive matched baseline.

Group-level ALC ranges from approximately 0.0306 to 0.2672 (Appendix D); these anonymized groups cannot be mapped reliably to named hidden families. The public R1 Brier of 0.153728 and the hosted ALC of 0.159061 are not comparable accuracy estimates because their questions and budget weighting differ. In particular, the available hosted feedback does not identify how much of its gain is due to recurrence rather than the initial forecasting context.

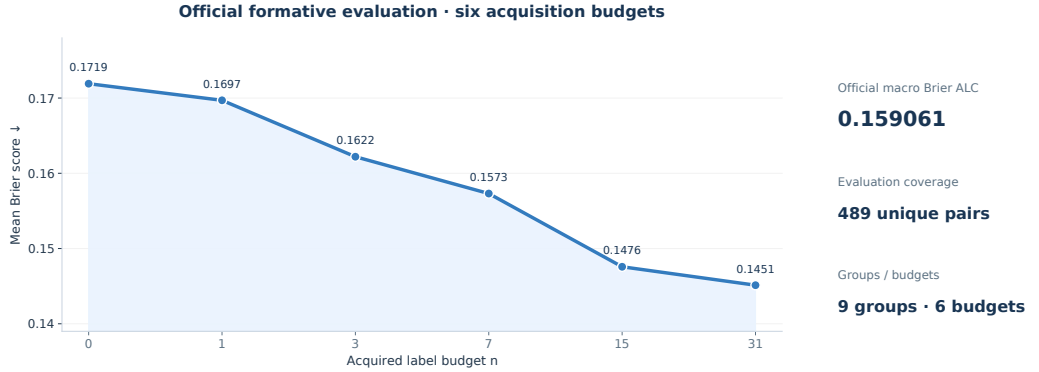


Figure 5. Official formative six-budget learning curve for BLE 1.5-A. Macro group Brier decreases across budgets 0, 1, 3, 7, 15, 31; the official Brier ALC is 0.1590607804. The same submission covers nine subject–benchmark groups and 489 distinct pairs. The plot does not isolate the causal gain from label acquisition.

6 Discussion, reliability and limitations

What the evidence supports. The frozen public comparison supports a paired reduction in diagnostic Brier for BLE R1 versus direct Luna on fresh questions in four previously studied families. The hosted result separately demonstrates a complete officially scored, six-budget Responses API deployment. Together they establish experimental promise and practical feasibility; neither establishes a general hidden-family gain over direct Luna or a causal effect of retrieval.

Reliability is part of the method. Explicit belief retention and trusted evidence controls matter operationally because the benchmark invokes a full executable pipeline. Local audit and mocked fault-injection evidence can identify uncaught exception paths, but cannot supply a statistical estimate of hosted failure probability. Throughput depends on model response latency, worker concurrency, provider throttling and cold retrieval-model initialization. Codabench's exposed job status does not establish the organizer-counted duration; compliance with the eight-hour rule cannot be certified from the score alone.

Negative transfer and methodological uncertainty. A separate rank-16 frozen-embedding system–item factor model performed well on some within-family held-out items but lost to a calibrated scalar baseline in both strict cross-family directions. This is a negative component experiment rather than a matched comparison with BLE (Appendix E). Similarly, the initial-to-final BLE comparison does not control deliberation time: a same-prompt, same-compute retrieval ablation

would be needed to identify the evidence contribution. Calibration, evidence relevance, benchmark dependence and unobserved provider runtime remain limitations.

Scope of later engineering tests. A configuration-only BLE 1.5-A.2 Fast candidate was separately packaged and submitted (#1033843). Codabench reports *Failed* with runtime diagnostic `PAIEC-UNKNOWN-001`. The cause, including whether the run timed out, is unverified. Its 3,000-forecast mocked fault campaign informs engineering coverage, not official accuracy or measured API reliability. No 1.5-A.2 result is included in the 1.5-A ALC reported here.

7 Conclusion

An audited recurrent linguistic evaluator outperformed direct Luna on a limited, outcome-sealed public panel, and a later Responses API implementation completed the official six-budget formative evaluation with **Brier ALC 0.1590607804**. These are distinct findings. The value of the approach lies in explicitly recording what the model believed, what permitted evidence it requested, and how its probability changed; this makes failure modes, calibration and transport validity inspectable. Future work should compare retrieval with matched deliberation and measure acquisition-policy value and hosted runtime without conflating them with the already successful formative score.

Attribution and AI assistance. The BLE architecture and baseline retrieval interface originate with AIMS Foundations [2]. GPT-6 Luna and ChatGPT/Codex assisted software development, experiment execution, analytical drafting and figures; the author is responsible for the verification, interpretation and claims. Diagrams are conceptual; numerical plots reproduce recorded values, not AI-invented observations.

References

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[4] G. W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

[5] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

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A Research protocol and source-group controls

A.1 Outcome-sealed confirmation sequence

The initial experiment compared R1 and R2 on 16 questions from four development families and budgets 0, 7, with 32 frozen A0 question–budget controls. R1 was promoted under the amended continuation rule. Its confirmation used 32 additional source-group-disjoint questions, eight per family, at $n = 0, 7$; outcomes were opened only after forecasting was frozen. A single question–budget row may summarize repeated observed binary executions. The archived scorer's source-group bootstrap protects against treating both budgets of the same item as independent observations. The four families were known during prior research, so source-group separation is not family separation.

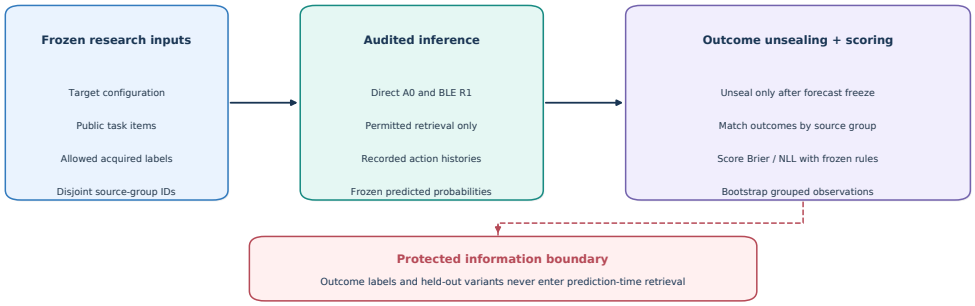


Figure 6. Outcome-sealed public research protocol. Public target inputs and allowed acquisition labels feed direct or recurrent predictions, with coordinator-enforced source exclusions. Forecasts are frozen before independently scored outcomes are opened. This is the research confirmation boundary, not the hosted private scorer's internal implementation.

Table 3. Previously revealed development diagnostics. Method-selection scores, not prospective evidence.

Method	Brier	NLL
Direct A0	0.224650	0.659165
R1 initial	0.228831	0.660757
R1 final	0.208044	0.613407
R2 initial	0.217958	0.620662
R2 final	0.209687	0.602757
Beta(4,4)	0.225926	0.642732
Frozen 50/50 hybrid	0.210304	0.610639

Table 4. Public confirmation by budget. Each budget shares the same underlying 32 questions.

Acquired n	Direct A0	Final R1	Gain (A0–R1)
0	0.239586	0.175754	+0.063832
7	0.168180	0.131702	+0.036479

Reported source-cluster bootstrap 95% intervals were [0.0015691295, 0.1030590290] for the total paired gain and [0.0008782087, 0.0449844010] for the initial-to-final R1 difference. They do not provide a family-superpopulation guarantee.

B Additional method illustrations

B.1 Explicit state register

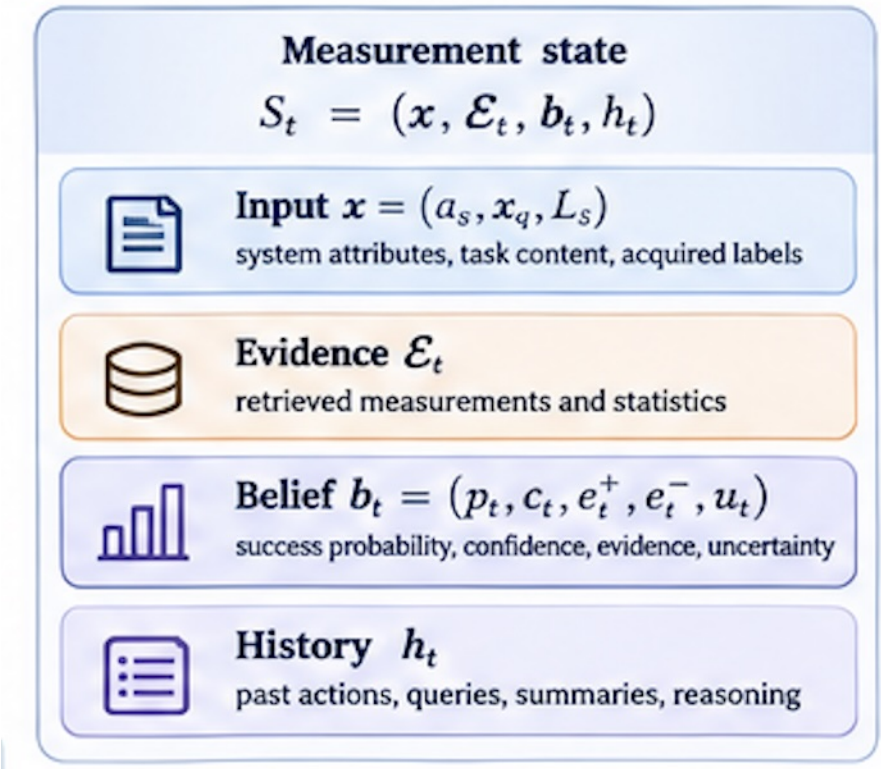


Figure 7. Explicit measurement state. Fixed inputs $x = (a_s, x_q, \mathcal{L}_s)$, admissible evidence $\mathcal{E}_t$, expressed belief b_t , and action history h_t comprise the auditable state S_t . It is a linguistic record, not a learned recurrent neural memory.

B.2 Coordinator-mediated action flow

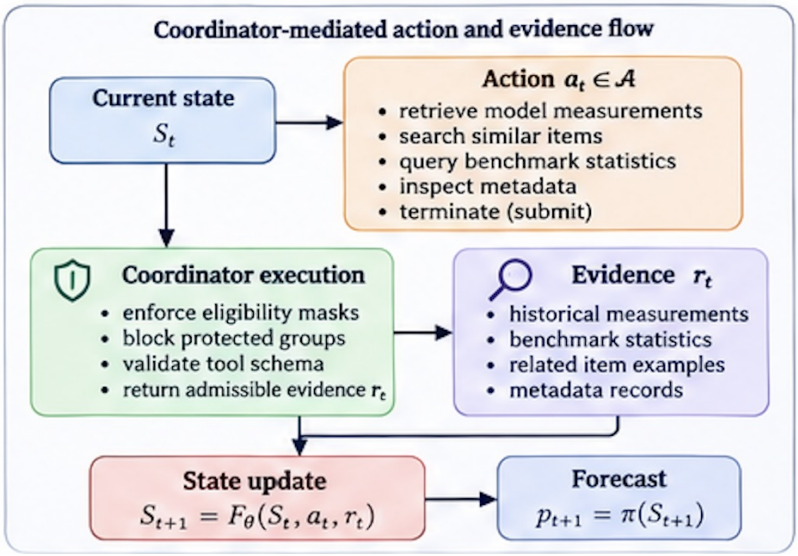


Figure 8. Coordinator-mediated belief update. The evaluator proposes an admissible action a_t based on state S_t . The coordinator validates its request and executes eligible retrieval, returning evidence r_t that conditions the next expressed belief. The schema and eligibility checks are deterministic; judging evidence relevance is model-driven.

B.3 Iterative belief trajectory

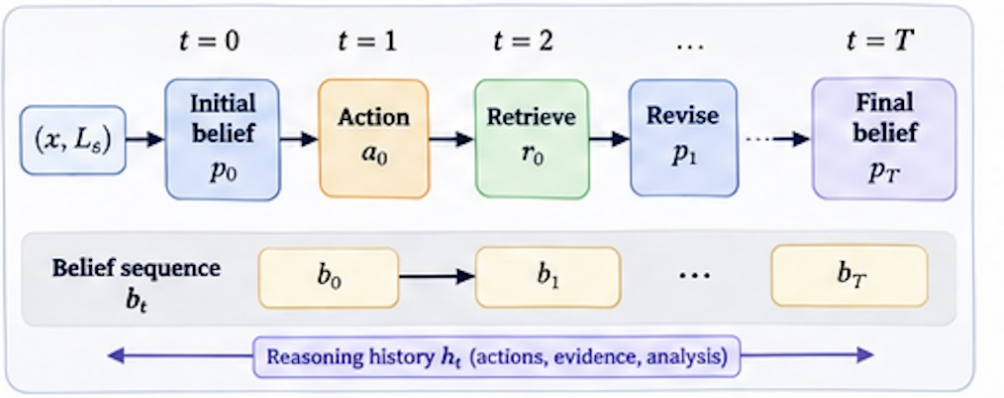


Figure 9. Iterative inference and belief revision. The bounded sequence of initial belief, admissible action, retrieved evidence, and revised belief carries an explicit probability and reasoning history. This is a schematic of possible steps, not an assertion that every evaluated forecast used all steps.

B.4 Brier-loss geometry

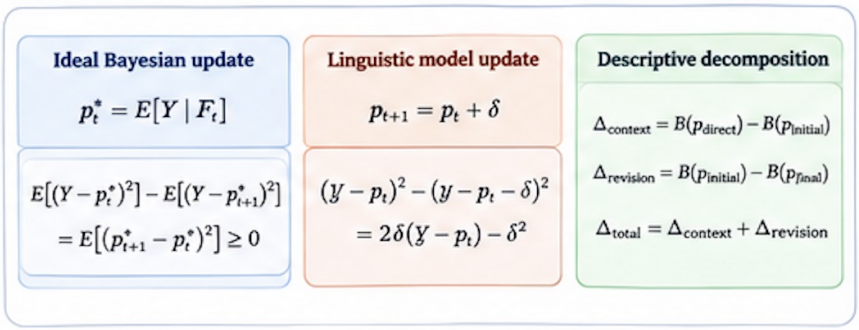


Figure 10. Ideal and approximate belief updates. Under nested information, an ideal conditional mean cannot worsen expected squared error; an approximate linguistic update may improve or harm realized Brier loss. The right panel decomposes the observed R1 advantage descriptively into initial-prompt context and subsequent revisions, without claiming causal retrieval effects.

C Public family heterogeneity

The evaluator changed its initial probability in 36 of 64 confirmation forecasts; 55 of 64 returned at least one quantitative evidence result. The audited coordinator executed 56 semantic item searches, 54 model lookups, 22 benchmark searches, 11 score lookups and 3 benchmark-detail queries. These counts show evidence use, not retrieval causality. Some revisions reflected task reasoning even when retrieved analogues were judged weak; other revisions referenced historical task scores while qualifying their transfer relevance.

Table 5. Diagnostic Brier by public confirmation family. Equal-budget average over 0 and 7. Gain is direct A0 minus final R1.

Family	Direct A0	Final R1	Gain
LexEval	0.225900	0.164944	+0.060956
LiveAoPSBench	0.305694	0.211300	+0.094394
MathArena	0.295094	0.237844	+0.057250
MedAgentBoard	0.034450	0.042881	−0.008431
Macro mean	**0.203883**	**0.153728**	**+0.050155**

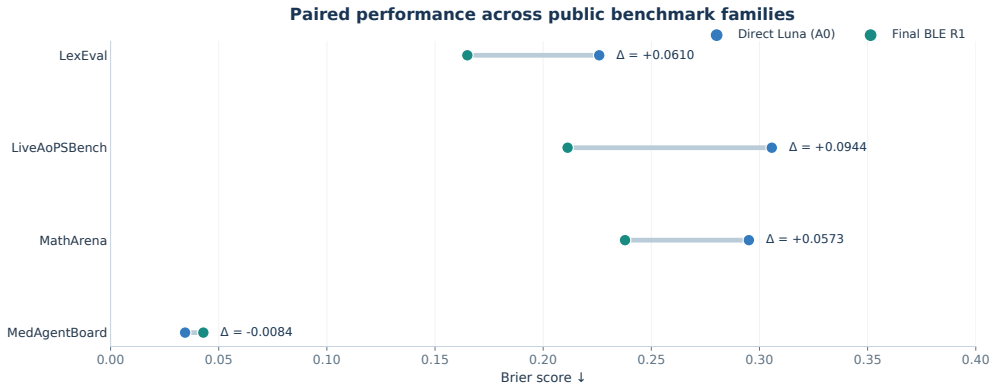


Figure 11. Family heterogeneity in public confirmation. Final R1 improves relative to direct Luna in three of four previously encountered benchmark families and slightly deteriorates on MedAgentBoard. This is not a cold-family evaluation.

D Hosted group heterogeneity and score reconciliation

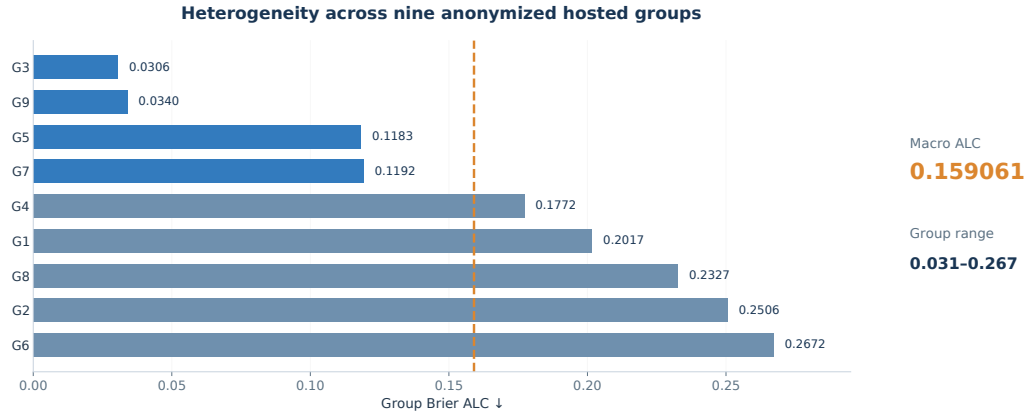


Figure 12. Anonymized hosted group heterogeneity. Official BLE 1.5-A formative group ALC values span approximately 0.0306–0.2672. The dashed line shows the unweighted mean over nine subject–benchmark groups, 0.159061. Display labels G1–G9 are arbitrary group indices and do not identify the underlying hidden benchmarks.

Table 6. Hosted ALC reconciliation. Group indices are the figure's labels, not private benchmark identities. The published scorer supplies more precision than the rounded six-decimal entries below.

Group	Pairs	ALC	Group	Pairs	ALC
G1	44	0.201726	G6	60	0.267202
G2	56	0.250581	G7	54	0.119215
G3	58	0.030571	G8	44	0.232692
G4	44	0.177229	G9	58	0.033989
G5	71	0.118341			

The nine pair counts sum to 489 and their equally weighted ALC values average to 0.1590607 after rounding. Weighting groups by their different pair counts would change the estimand and must not be substituted for the official group macro average.

E Inductive factor-model negative result

We separately investigated a frozen 768-dimensional EmbeddingGemma 2 item representation with a trainable rank-16 system–item interaction model,

$$\hat{p}_{s,q} = \sigma\big(b + b_s + u_s^{\mathsf{T}} W \phi(q)\big). \tag{8}$$

The experiment used 694 items, 61 configurations, and 25,992 observations from AlgoTune and EDIT-Bench. Although within-family held-out items sometimes benefited from the factor representation, the seven-label calibrated source-only transfer deteriorated in both directions. This is not a paired evaluation against BLE on identical examples. Mean NLL worsened from 0.628290 to 0.675636.

Table 7. Grouped held-out-item factor diagnostics. Privileged oracle is diagnostic only.

Family	Scalar	Factor	Oracle
AlgoTune	0.195783	0.258331	0.152869
EDIT-Bench	0.243105	0.159346	0.139571
Equal-family mean	0.219444	0.208838	0.146220

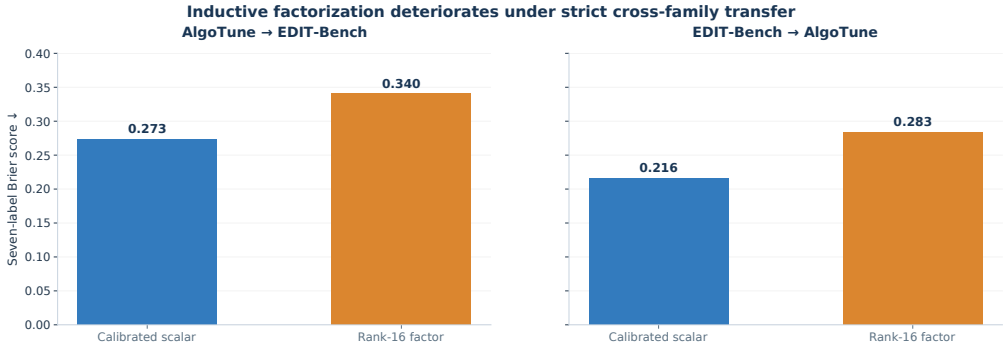


Figure 13. Strict cross-family transfer failure. At seven labels, the rank-16 factor predictor scores 0.340420 versus scalar 0.272994 for AlgoTune→EDIT-Bench, and 0.283372 versus scalar 0.216085 for the reverse direction. Lower Brier is better.

The embedding pass took 16.61 minutes; 13 long items required chunk pooling. Seven numerical fits were approximately three seconds once embeddings were available. Residual-ranking Spearman under strict transfer was only 0.0404 and 0.0249, supporting retention of the negative finding rather than scaling the model.

F Runtime reliability, stress tests and reproducibility

F.1 Failure containment is not finalization

When a model returns a valid initial or intermediate probability, the coordinator may retain it if a later request is interrupted before the intended evidence trajectory completes. This is a model-derived belief, not a constant-probability emergency substitution. Nevertheless, a valid fallback belief is distinct from a fully completed revision. A configuration-only A.2 Fast release was tested with 3,000 mocked forecasts, 200 injected faults, 9,020 mocked API requests, and zero uncaught batch errors or emergency substitutions. Six budgets and 63 exact archived retrieval packets were exercised. The experiment did not measure paid API forecast accuracy, hosted runtime or a real-world fault rate. In a separate synchronized 16-worker test, four of 48 forecasts retained prior valid beliefs at the original deadline despite all workers completing local retrieval; this is a qualified test rather than perfect evidence completion.

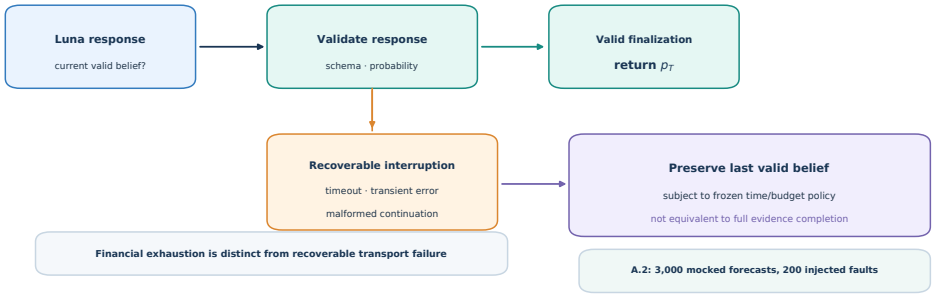


Figure 14. Schematic of bounded inference and valid-belief retention. An accepted final forecast is distinguished from a recoverable interruption that leaves an earlier model-derived probability available. A.2 mocked stress tests probe this logic but do not prove hosted API failure probabilities or eight-hour execution compliance.

F.2 Research versus production provenance

The historical audited-retrieval archive `BLE_AUDITED_RETRIEVAL_002_R2_RESULTS.zip` contains the frozen `TRANSPORT-R2/confirmation/results.json`, scored rows, source-group masks, exclusion and failure records. Its recorded SHA-256 is:

`4be365d4b9db29f93ae09276da5b1c2fcaf70470e031ddd7374a2abdb2d4e708`

Archived verification reports 6,538 individual file hashes and a passing ZIP CRC. The public research transport executed 192 successful forecasting sessions across development and confirmation, with a median successful forecast latency of 29.9 seconds and 95th percentile 55.1 seconds in the confirmation study. The associated Codex/ChatGPT token counts are not portable measures of paid Responses API billing or hosted execution time.

The successfully scored production release is `PAIEC_BLE15_A_PRODUCTION_001.zip`, submitted as #1032237. Its recorded official score is 0.1590607804. The verified SHA-256 of the local frozen production archive is:

`90a6a22560328e4ff5ec3f86375dbf567aecf802d2632fe6829878b339be7c3c`

Exact matching of public-safe source to the submitted artifact remains a release task. The A.2 candidate (#1033843) was a separate Fast-tier experiment and received no score credited to this result.

F.3 Public code and style compliance

The designated release location is https://github.com/KarimGhandour/PAIEC-code-submission. Before public submission, that repository must be populated and verified with a sanitized source tree, pinned dependencies, required licensed assets or acquisition instructions, exact release mapping, and rerun documentation. No embedded API credential or hidden evaluation data may enter the public repository. Full release verification is a pending deliverable, not an accomplished result stated by this manuscript.

This manuscript uses the official NeurIPS 2026 stylesheet in attributed preprint mode. The illustrations were recovered from the approved manuscript PDF, and the reconstruction record documents their provenance.